# Exercise Sheet 5

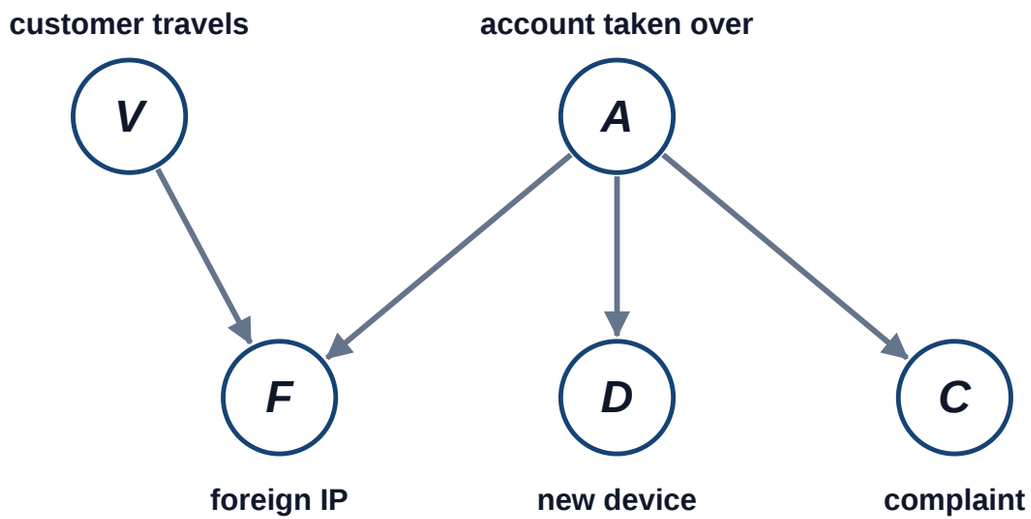
## Bayesian Networks

Probabilistic Machine Learning

Prof. Dr. Michael Granitzer

# 5.A Graphing a Joint

# Exercise 5.A.1: A Small Fraud Network




All five variables are yes/no.

1. Write the joint $p(V, A, F, D, C)$ as the network factorises it.
2. How many numbers does the network need? How many the full joint?
3. State the local Markov claim for $D$ and for $F$.
4. Name one missing arrow you would challenge, and what adding it costs.

# Exercise 5.A.1: A Small Fraud Network — Solution

# Exercise 5.A.1: A Small Fraud Network — Solution

**1.** $p(V, A, F, D, C) = p(V) \cdot p(A) \cdot p(F \mid V, A) \cdot p(D \mid A) \cdot p(C \mid A)$.

**2.** $1 + 1 + 4 + 2 + 2 = 10$ numbers, against $2^5 - 1 = 31$.

**3.** $D$ has parent $A$ and no descendants: $D \perp\!\!\!\perp \{V, F, C\} \mid A$. $F$ has parents $V$, $A$: $F \perp\!\!\!\perp \{D, C\} \mid V, A$.

**4.** For example $V \to D$: travellers often use a new phone or a hotel computer. It removes the claim $D \perp\!\!\!\perp V \mid A$ and turns $p(D \mid A)$ into $p(D \mid V, A)$: 4 numbers instead of 2, 12 in total. (Any well-argued arrow counts; the point is that each missing arrow is a claim.)

# Exercise 5.A.2: Inference by Elimination

Tables: $P(V{=}1) = 0.05$, $P(A{=}1) = 0.01$; $P(F{=}1 \mid V, A)$: $0.02$ for $(0,0)$, $0.9$ for $(1,0)$, $0.7$ for $(0,1)$, $0.95$ for $(1,1)$; $P(D{=}1 \mid A)$: $0.1$ for $A{=}0$, $0.6$ for $A{=}1$; $P(C{=}1 \mid A)$: $0.001$ for $A{=}0$, $0.5$ for $A{=}1$.

1. Compute $P(F{=}1)$. Which variables can be dropped before summing, and why? Eliminate $V$ first and give the message $m(A)$.
2. Compute $P(A{=}1 \mid F{=}1)$.
3. Compute $P(A{=}1 \mid F{=}1, D{=}1)$.
4. Compute $P(A{=}1 \mid F{=}1, V{=}1)$ and compare with 2. Name the effect.

# Exercise 5.A.2: Inference by Elimination — Solution

# Exercise 5.A.2: Inference by Elimination — Solution

**1.** $D$ and $C$ are unobserved leaves: $\sum_d p(d \mid A) = 1$ and $\sum_c p(c \mid A) = 1$, so they drop out. Then $P(F{=}1) = \sum_A p(A) m(A)$ with $m(A) = \sum_V p(V) P(F{=}1 \mid V, A)$: $m(0) = 0.95 \cdot 0.02 + 0.05 \cdot 0.9 = 0.064$, $m(1) = 0.95 \cdot 0.7 + 0.05 \cdot 0.95 = 0.7125$. $P(F{=}1) = 0.99 \cdot 0.064 + 0.01 \cdot 0.7125 = 0.06336 + 0.007125 = 0.070485$.

**2.** $P(A{=}1 \mid F{=}1) = 0.007125 / 0.070485 \approx 0.101$ — ten times the prior.

**3.** Multiply in $P(D{=}1 \mid A)$: $0.007125 \cdot 0.6 = 0.004275$ and $0.06336 \cdot 0.1 = 0.006336$, so $P(A{=}1 \mid F{=}1, D{=}1) = 0.004275 / 0.010611 \approx 0.403$.

**4.** $P(A{=}1, V{=}1, F{=}1) = 0.01 \cdot 0.05 \cdot 0.95 = 0.000475$ and $P(A{=}0, V{=}1, F{=}1) = 0.99 \cdot 0.05 \cdot 0.9 = 0.04455$, so $P(A{=}1 \mid F{=}1, V{=}1) = 0.000475 / 0.045025 \approx 0.011$ — back to about the prior. Travel **explains away** the foreign IP: $V$ and $A$ are independent, but not given their common effect $F$.

# 5.B Reading Independence

# Exercise 5.B.1: d-Separation

For the fraud network of 5.A.1, decide with d-separation — give the path and the triple that blocks or opens it:

1. $V \perp\!\!\!\perp A$ ?
2. $V \perp\!\!\!\perp A \mid F$ ?
3. $V \perp\!\!\!\perp D$ ?
4. $V \perp\!\!\!\perp D \mid F$ ?
5. $V \perp\!\!\!\perp D \mid F, A$ ?
6. $D \perp\!\!\!\perp C$ ? and $D \perp\!\!\!\perp C \mid A$ ?

Then check 3 and 4 with the numbers of 5.A.2: compare $P(V=1 \mid F=1)$ with $P(V=1 \mid F=1, D=1)$.

# Exercise 5.B.1: d-Separation — Solution

# Exercise 5.B.1: d-Separation — Solution

| query | path | verdict |
|---|---|---|
| $V \perp\!\!\!\perp A$ | $V \to F \leftarrow A$: collider, unobserved | **yes** |
| $V \perp\!\!\!\perp A \mid F$ | the collider is observed: open | **no** |
| $V \perp\!\!\!\perp D$ | $V \to F \leftarrow A \to D$: collider $F$ unobserved | **yes** |
| $V \perp\!\!\!\perp D \mid F$ | collider $F$ open, fork $A$ unobserved: open | **no** |
| $V \perp\!\!\!\perp D \mid F, A$ | fork $A$ observed: blocked | **yes** |
| $D \perp\!\!\!\perp C$ | $D \leftarrow A \to C$: fork, unobserved | **no** |
| $D \perp\!\!\!\perp C \mid A$ | fork observed | **yes** |

**Numbers.** $P(V{=}1) = 0.05 = P(V{=}1 \mid D{=}1)$ (claim 3). Given $F{=}1$: $P(V{=}1 \mid F{=}1) = 0.045025/0.070485 \approx 0.639$, but $P(V{=}1 \mid F{=}1, D{=}1) = 0.00474/0.010611 \approx 0.447$: a new device makes a take-over more likely, which explains the foreign IP and makes travel less likely. Claim 4 fails, as d-separation said.

# Exercise 5.B.2: Which Arrows Are in the Data?

1. Which of these graphs are Markov equivalent? (a) $X \to Y \to Z$ · (b) $X \leftarrow Y \leftarrow Z$ · (c) $X \leftarrow Y \to Z$ · (d) $X \to Y \leftarrow Z$ · (e) (a) plus $X \to Z$
2. In the fraud network of 5.A.1, which single arrows can be reversed without changing any independence claim? Can $A \to D$ and $A \to C$ be reversed together?
3. An analyst fits the fraud network from data and says "take-overs **cause** new devices". What can the data support?

# Exercise 5.B.2: Which Arrows Are in the Data? — Solution

# Exercise 5.B.2: Which Arrows Are in the Data? — Solution

**1.** (a), (b), (c) share the skeleton $X - Y - Z$ and have no collider: all claim $X \perp\!\!\!\perp Z \mid Y$ and nothing else — **equivalent**. (d) has the collider at $Y$: it claims $X \perp\!\!\!\perp Z$ instead — different. (e) is complete and claims no independence at all — different from all four.

**2.** Equivalent graphs need the same skeleton and the same colliders. The collider $V \to F \leftarrow A$ must stay, so $V \to F$ and $A \to F$ cannot be reversed. $A \to D$ alone can be reversed ($D \to A \to C$ creates no collider), and so can $A \to C$ alone. Both together give $D \to A \leftarrow C$, a new collider — not equivalent.

**3.** Only that $A$ and $D$ are dependent, and — since $A \to D$ is reversible — not its direction. The causal reading comes from knowledge outside the data (a take-over happens before the device is used), not from the fit.

# 5.C Plates and Learning

# Exercise 5.C.1: Naive Bayes as a Plate Diagram

Naive Bayes for $N$ transactions with $p$ yes/no flags: class prior $\pi = P(y = 1)$ with a $\mathrm{Beta}(a, b)$ prior; flag probabilities $\theta_{jc} = P(x_j = 1 \mid y = c)$, each with a $\mathrm{Beta}(1, 1)$ prior.

1. Draw the plate diagram: nodes, shading, dots, plates (and which plates are nested).
2. Write the factorisation that the diagram encodes.
3. At prediction time a new transaction arrives with flags but no label. What changes in the drawing?

# Exercise 5.C.1: Naive Bayes as a Plate Diagram — Solution

# Exercise 5.C.1: Naive Bayes as a Plate Diagram — Solution

**1.** Dot $(a, b)$ → open node $\pi$ → shaded $y_i$. A plate over $N$ transactions holds $y_i$ and, **nested** inside it, a plate over the $p$ flags holding the shaded $x_{ij}$. Outside the $N$ plate: open nodes $\theta_{jc}$ in a plate over the $p$ flags and a plate over the 2 classes, fed by a dot $(1, 1)$. Arrows $y_i \to x_{ij}$ and $\theta_{jc} \to x_{ij}$ ($y_i$ selects which $\theta_{j\cdot}$ applies).

**2.**

$$p(\pi \mid a, b) \prod_{j=1}^{p} \prod_{c=0}^{1} p(\theta_{jc}) \prod_{i=1}^{N} \left[ p(y_i \mid \pi) \prod_{j=1}^{p} p(x_{ij} \mid y_i, \theta_{j0}, \theta_{j1}) \right]$$

**3.** The new label $y_{\text{new}}$ is **unshaded** — it is inferred. The parameters are either conditioned on the training data (the full Bayesian answer) or replaced by their smoothed counts — the posterior means of Lecture 03.

# Exercise 5.C.2: Counting, Elimination, and an Empty Row

1. A chain $X_1 \to \cdots \to X_{10}$ of yes/no variables. How many terms does $p(x_{10})$ need if all other variables are summed at once, and roughly how many operations by elimination?
2. From 200 records of the fraud network (table), estimate $P(F{=}1 \mid V, A)$ by maximum likelihood and with one pseudo-count per value. What happens to the last row, and what does smoothing put there?

| $V$ | $A$ | records | with $F = 1$ |
|---|---|---|---|
| 0 | 0 | 180 | 4 |
| 1 | 0 | 15 | 13 |
| 0 | 1 | 5 | 3 |
| 1 | 1 | 0 | 0 |

# Exercise 5.C.2: Counting, Elimination, and an Empty Row — Solution

# Exercise 5.C.2: Counting, Elimination, and an Empty Row — Solution

**1.** $2^9 = 512$ terms (each a product of 10 factors), against about $4 \cdot 9 = 36$ multiplications by passing a two-entry message along the chain.

**2.**

| $V, A$ | MLE | smoothed $(n_1 + 1)/(n + 2)$ |
|---|---|---|
| $0, 0$ | $4/180 \approx 0.022$ | $5/182 \approx 0.027$ |
| $1, 0$ | $13/15 \approx 0.867$ | $14/17 \approx 0.824$ |
| $0, 1$ | $3/5 = 0.6$ | $4/7 \approx 0.571$ |
| $1, 1$ | $0/0$ — undefined | $1/2 = 0.5$ |

A parent configuration that never occurred leaves its row **undetermined** under maximum likelihood; the pseudo-counts fill it with the prior mean $1/2$. Sparse rows are one more reason to keep the number of parents small.